

controlling the parameters

- Max current 2A
- Monitors from 1S battery cell
- Windows 10 compatible

software

- I2C serial interface
- State of charge measurement

Safety features.

- Charger detection
- JEITA charge protection
- Thermal protection
- Battery cell monitoring
- Open circuit voltage
- Temperature monitoring
- Current monitoring

The reference design can be used for various applications, which include portable consumer devices, digital imaging devices (DSC, DVC), portable industrial devices, USB 1-cell pack equipment, etc. More details about the reference design can be viewed by scanning the QR code.



DA9318 - 10A fast battery charger

DA9318 from Renesas is a Li-ion fast cell charger reference design that can be used as a standalone 1S charger or can be implemented into laptops, smartphones, tablets, etc. The charger module employs an ATMEL SAM3U microcontroller as the USB transceiver. This adds the I2C functionality to the board and also the discrete digital IO control GPIO and dedicated functions.

The charge management IC DA9318M is capable of providing a maximum charging current of 10A. The charger is capable of charging with up to 98% efficiency delivering 2A charge current. This charger employs a high-efficiency current doubler with an integrated ADC for system monitoring.

The reference design is capable of handling current up to 10A while maintaining the safety of the

battery. It operates together with the main charger, which handles the pre-charge and constant voltage charging duties. The charger is capable of delivering up to 6A current by using the standard Type C cable. The fast charging is achieved by the doubler circuit that uses a high-efficiency capacitive divider. The output voltage it provides is given by $V_{IN}/2$.

The module has multiple safety features, which include an integrated reverse protection that blocks current flow in both the directions when the device is in off state. The DA9318 employs six hardware based safety functions for any undervoltage or overvoltage condition. All safety-triggered events lead to an automatic shutdown and are reported via an interrupt to the system.

It employs an 8-bit ADC to keep a check on parameters such as input current, output current, voltage, and junction temperature. For software supervision a programmable watchdog timer and for battery overload protection a safety timer are included.

The module employs two power monitoring INA226 ICs, which enable the measurement of the main charger input voltage and input current and allows the monitoring of the battery parameters. I2C protocol is used for communication.

When the charge current reaches a handover charge current threshold, which is arbitrarily set to 0.9A, the DA9318 is switched off and the main charger is enabled and finishes off the charge cycle. The module is over 98% efficient when supplying an output current of 0.9A.

Important components.

U1 - DA9318 charging IC

U2 - ATMEL SAM3U microcontroller

Features.

- Safe high-voltage direct charging
- Travel adaptor detection

- User programmable overcurrent protection threshold
- Built-in fault protection
- High immunity of false triggering under transients
- Responds to input overvoltage in less than 1µs
- Fault indication for various fault occurrences
- 10A output current (DA9318M)
- 98% efficiency at 2A
- 10% current sense accuracy (DA9318M)

- Reverse and forward current protection in IDLE mode
- I2C compatible 2-wire interface

Safety features.

- Automatic shutdown in fault condition
- High-accuracy protection thresholds
- Overcurrent protection
- Reverse and forward current protection in IDLE mode
- Junction temperature monitoring

This charger reference design can be used in various devices, including mobile, standalone Li-ion battery chargers, tablets, larger battery packs, laptops, etc. More details about the reference design can be found by scanning the QR code. **EFY**



EFY note. To read more about each of the above reference designs please visit electronicforu.com or scan the QR code given below.



The author, Sharad Bhowmick, works as a Technology Journalist at EFY. He is passionate about power electronics and energy storage technologies. He wants to help achieve the goal of a carbon neutral world.